INCH-POUND
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MILITARY SPECIFICATION

GENERATOR SET, DIESEL ENGINE DIRECT AND ALTERNATING CURRENT (NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers constant speed, continuous duty direct current (dc) and alternating current (ac) diesel engine generator sets.

1.2 Classification. Generator sets are of the following types, as specified (see 6.2):

Type I - DC, single engine driven for surface ships.
Type II - AC, single engine driven for surface ships.
Type III - AC, twin engine driven for surface ships.
Type IV - AC, single engine driven for submarines.

1.3 Service. Generator sets are furnished to provide one of the following service categories, as specified (see 6.2):

Ship service - Prime electric power required to power all essential loads and non-essential loads.

Emergency service - Secondary electric power required for all essential loads in the event of loss of ship service electric power.

Ship service/emergency service - Sets which are used to provide both ship service and emergency service.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

ZZ-R-765 - Rubber for Mountings (Unbounded-Spool and Compression Types).
MMM-A-260 - Adhesive, Water-Resistant (for Sealing Waterproofed Paper).
MMM-A-1617 - Adhesive, Rubber Base, General Purpose.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of
MIL-B-131 - Barrier Materials, Watervaporproof, Greaseproof, Flexible, Heat-Sealable.
MIL-S-19 - Support Items, Accessories and Kits, Mechanical; Packaging of
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-C-2212 - Contactors and Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-R-2178 - Regulator Sets, Voltage, A.C. Generator, Naval Shipboard Use.
MIL-G-3111 - Generators, Electric, Direct-Current (Naval Shipboard Use).
MIL-G-3174 - Generator, Alternating Current, 60-hertz (Naval Shipboard Use).
MIL-C-9656 - Container, Shipping and Storage (Flexible, Reusable, Water-Vaporproof).
MIL-E-10097 - Engines; Preparation for Shipment and Storage of
MIL-G-17803 - Gasket Material, Non-Metallic.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment
MIL-P-15024/3 - Plates, Identification

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MILITARY (Continued)

MIL-T-15108 - Transformers, Power, Step-Down, Single-Phase, 60-Hertz, 1-Kilovoltampere Approximate Minimum Rating, Dry Type, Naval Shipboard.
MIL-R-15109 - Resistors and Rheostats, Naval Shipboard.
MIL-S-15371 - Starter, Engine; Electrical Motor (Shipboard Use).
MIL-S-16036 - Switchgear, Power, Naval Shipboard.
MIL-E-16298 - Electric Machines Having Rotating Parts, Accessories and Associated Support Items: Packaging of.
MIL-M-17059 - Motors, 60-Cycle, Alternating Current, Fractional H.P. (Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts). Packaging of.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-C-19836 - Coolers, Fluid, Industrial, Air Motor and Generator, Naval Shipboard.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-21410 - Governing Systems, Speed and Load-Sensing (for Electric Generator Sets).
MIL-E-23457 - Engines, Diesel Marine, Propulsion and Auxiliary, Medium Speed.
MIL-I-24453 - Inhibitor, Corrosion, Soluble-Oil.
MIL-E-24455 - Engines, Diesel, Propulsion and Auxiliary, High Speed, Naval Shipboard.
MIL-I-26860 - Indicator, Humidity, Plug, Color Change.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-167-2 - Mechanical Vibrations of Shipboard Equipment (Reciprocating Machinery and Propulsion System and Shafting) Types III, IV and V.
MIL-STD-178 - Definitions Applicable to Speed-Governing of Electric Generator Sets.
MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MILITARY (Continued)

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-756 - Reliability Modeling and Prediction.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-882 - System Safety Program Requirements.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety.
MIL-STD-1629 - Procedures for Performing a Failure Mode Effects and Criticality Analysis.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
S9073-A2-HBK-010 - US Navy Resilient Mount Handbook.
S9086-HD-STM-000 - Diesel Engines Propulsion and General Purpose, Naval Ships' Technical Manual, Chapter 235.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Materials. Materials shall be in accordance with MIL-E-917.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Nonstructural parts. Nonstructural parts such as terminal boxes, housings, ducts, and similar parts shall be fabricated from aluminum alloy, where feasible.

3.2.3 Hazardous materials. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the Bureau of Medicine (BUMED) who will act as an advisor to the contracting activity. Materials and parts containing radioactive material, mercury, cadmium, asbestos, carbon, boron fibres, or other materials giving off toxic fumes under operating or casualty conditions shall not be used.

3.2.3.1 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.8).

3.3 Diesel generator set.

3.3.1 General requirements. The diesel generator set and accessories shall be as lightweight and compact as possible consistent with the construction, performance, reliability and maintainability requirements specified herein (see 6.3 and appendix B). Weight and dimensional limitations shall be as specified (see 6.2).

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3.3.1.1 Construction. The requirements specified shall be considered as minimum standards and the construction shall have as primary objectives the following:

(a) Maximum reliability.
(b) Maximum ease of operation.
(c) Maximum maintainability.

3.3.1.2 Reliability and maintainability. The contractor shall employ in the process of design and manufacture such methods as are necessary to assure achievement of levels of reliability and maintainability of the complete generator set not less than those specified in 3.3.1.2.1, 3.3.1.2.3 and 3.3.1.2.5 (see 6.3).

3.3.1.2.1 Life. The diesel generator set design and construction shall be based on a life expectancy as:

Ship service and emergency service - 125,000 hours of operation with no limitations as to the number of starts and stops. This includes 100,000 hours at rated load, 20,000 hours between 50 percent and 100 percent rated load, and 5,000 hours below 50 percent rated load. The time between overhauls shall be not less than 20,000 hours.

3.3.1.2.2 Repair of parts. Parts which are not classified as on board parts shall not need repair, except during overhaul periods, or where damage is incurred due to external causes unrelated to the design.

3.3.1.2.3 Reliability prediction. The reliability prediction of the generator set shall be based on the failure rate of parts, and the information specified in 3.3.1.2.1 and 3.3.1.2.2. The construction and application of parts shall be such that the Mean-Time-Between-Failure (MTBF) of the generator set is at least 1,000 hours. The MTBF shall be predicted using the design prediction procedure specified in MIL-STD-756 and using the failure rate sources specified therein. For parts where no failure rate is available, a failure rate shall be estimated and the basis for the estimate stated in the calculations. The derating factors used for each part and the stress levels at which each part is operating shall be stated in reliability calculations. The stress levels shall be those as determined from test insofar as practicable. In those instances where test data is not available, design data shall be used.

3.3.1.2.4 Failure. A failure is defined as occurring when the collective or individual unit performance degrades to the point where the requirements of this specification cannot be met or wherein personnel safety is jeopardized.

3.3.1.2.5 Maintainability prediction. The Mean-Time-To-Repair (MTTR) shall be not more than 24 hours (see 6.3). The failure rates used shall be those which have been determined in the reliability predictions. The maintainability prediction shall take into consideration all design changes made during the engineering design phase.

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3.3.1.2.6 Failure mode effect and criticality analysis (FMECA). The contractor shall conduct, before the design tests, an FMECA in accordance with MIL-STD-1629 (see 6.3).

3.3.1.2.7 Maintenance. Accessibility shall be provided to all parts which require inspection, preventative maintenance, or replacement in service with the minimum practicable need for disconnection or removal of another part or assembly. Access panels, covers, peep holes, and so forth, shall be employed to the fullest extent feasible and compatible with the intended inspection or maintenance action. The envelope of the minimum required space around the generator set which is needed for ease of maintenance shall be as shown on the generator set installation drawing.

3.3.1.3 System safety program. The contractor shall implement a tailored system safety program (SSP) in accordance with task 100 (SSP), task 202 (Preliminary Hazard Analysis), and task 209 (Safety Assessment) of MIL-STD-882 (see 6.3). Safety design features, including fail-safe features, shall be incorporated into the design to prevent damage to equipment and to ensure optimal personnel protection during repair or interchanging of any component or assembly.

3.3.1.3.1 Safety analyses. Safety analyses shall be performed to identify hazardous conditions for the purpose of their elimination or control. Safety analyses shall examine the equipment, subsystem, components and their inter-relationship including logistic support, training, maintenance, and operational environment.

3.3.1.4 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.3.1.5 Generator set mounting. Generator sets in which the generator is not mounted off the engine crankcase shall be either sub-base or ship foundation mounted, as specified (see 6.2). Types I, II and IV generator sets may be either sub-base mounted or ship foundation mounted. Type III shall be sub-base mounted only. The generator rotor shall be directly connected to and in line with the engine crankshaft. The use of gear, belt or chain drives for the generator shall be prohibited.

3.3.1.5.1 Sub-base mounted. Sub-base mounted generator sets shall consist of a prime mover and generator and the attached accessories assembled and aligned as a unit on a welded steel sub-base which forms a rigid self-supporting structural unit for installation on its foundation in the ship without further stiffening or bracing.

3.3.1.5.2 Ship foundation mounted. Ship foundation mounted generator sets shall consist of the prime mover and generator each mounted directly to a ship-board foundation and aligned to form a generator set.

3.3.1.5.3 Single-bearing generators. Sets furnished with single-bearing generators shall have the engine end of the generator shaft piloted in either the engine flywheel or the engine crankshaft.

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3.3.1.5.4 Double-bearing generators. Sets furnished with double-bearing generators shall be coupled to the engine through a flexible coupling which absorbs slight misalignments.

3.3.1.5.5 Sound isolation - resilient mounts. When resilient mounts are specified (see 6.2), only Navy standard resilient mounts shall be used, as specified in S9073-A2-HBK-010. Mounts fabricated according to ZZ-R-768 are not acceptable. When resilient mounts are used (see 6.6), all mounts shall be of the same type and shall have equal loading. Mount attachment points shall be located so that the vertical distance between the horizontal center lines of mounts and the center-of-gravity of equipment is a minimum.

3.3.1.6 Couplings. Single-bearing generators larger than 60 kilowatts (kW) shall be provided with a forged-on half-coupling to suit the prime mover. Generators rated 60 kW and below may be provided with a shaft extension having a keyway and a shrunk-on coupling.

3.3.1.7 Generator bearing. Generator bearings shall be either sleeve or ball bearing on generators 40 kW and smaller having a flexible coupling between the prime mover and generator. Sleeve bearings shall be furnished on generators over 40 kW having a flexible coupling between the prime mover and generator. Generators with rigid coupling between the prime mover and generator, shall be furnished with sleeve bearings. The sleeve type bearing, when used, shall be force lubricated from the engine lubrication system and shall be provided with sight flow fittings at the bearing oil discharge. All generator bearing designs shall prevent current flow through the bearings caused by induced shaft voltages. The bearing insulation system shall be accomplished either by insulating the back side of the bearing or insulating a ring that the bearing rests on. The insulation system shall have a minimum number of pieces.

3.3.1.8 Generator frames. The frame shall have feet of ample size, large enough to accommodate holding-down bolts, dowel pins, and jackscrews where used, to ensure attachment to the sub-base. The frame of the generator shall not extend below the lower face of the sub-base.

3.3.1.9 Welding. Welding and allied processes shall be in accordance with MIL-STD-278.

3.3.1.10 Lifting means. Eyebolts, lugs, holes, or equivalent means shall be provided to permit lifting the generator set and attached accessories as a complete unit. Eyebolts or other means of lifting the generator alone shall not be used for lifting the assembled generator set.

3.3.1.11 Diesel engines. Diesel engines shall conform to MIL-E-23457 or MIL-E-24455 as specified (see 6.2). Unless otherwise specified (see 6.2), the direction of engine rotation shall be counterclockwise as viewed from the coupling end and shall be indicated by means of a metal arrow securely fastened to the engine block at the coupling end. Specified direction of rotation shall not apply to type III sets.

3.3.1.12 Grounding. The power and starting circuits shall be ungrounded. Exposed parts of electrical equipment (switches, meters, and so forth) shall be grounded. Bonding and grounding shall conform to MIL-STD-1310.

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3.3.1.13 Starting.

3.3.1.13.1 Starting equipment. Starting of the generator set shall be accomplished either by an electric starting or by a compressed air starting system as specified (see 6.2). Units over 100 kW shall be air start. Operation of starting system shall be automatic, semi-automatic, manual, or in combination as specified (see 6.2). For type IV generator set, only compressed air starting system shall be used.

3.3.1.13.1.1 Electric starting. Electric starting systems shall consist of an electric starting motor and motor starter. Starting motors shall conform to MIL-S-15371. The voltage and the degree of enclosure for the motor shall be as specified (see 6.2). Motor controllers shall conform to MIL-C-2212.

3.3.1.13.1.2 Compressed air starting. The compressed air starting system shall consist of an air starting motor, or distributor, applicable valves, strainer, lubricator, lines, connections and fittings. The ship's air supply shall be at pressure specified (see 6.2). The system shall connect to ship's air supply through an air pressure reducing valve for reducing high air pressure to engine air starting pressure. The system shall start the diesel engine automatically upon the failure of ship service power or the drop in ship service voltage below 65 percent of its rated voltage, and then cut off the air supply after the engine has started. A test switch located at the engine controls shall start the engine manually. A manually operated shutoff valve shall prevent inadvertent cranking of the engine when maintenance is being performed.

3.3.1.13.2 Starting methods. Engine generator sets shall incorporate starting methods as follows.

Ship service sets - normal and maintenance starts.

Emergency sets - emergency and maintenance starts.

Ship service/emergency sets - normal, emergency, and maintenance starts.

3.3.1.13.2.1 Engine generator starts. The engine generator starts referred to in 3.3.1.13.2 shall be defined as follows:

Normal start - Normal start is an operator initiated start request which provides for engine start up to idle speed where it remains until engine is properly warmed. The unit is brought up to operating speed either manually or by automatic paralleling provisions, if incorporated, in the ship installation.

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Emergency start - Emergency start is a start which is either operator initiated or automatic upon loss of voltage at the diesel generator switch board. Upon initiation of start sequence the generator set shall start, continue operating, and shall provide rated voltage at the generator terminals within 10 seconds for an ac generator set (30 seconds for a dc generator set) without the use of cold starting aids. The set shall provide full output power immediately after completion of the start sequence.

Maintenance start - A maintenance start is a start which brings the diesel to idle speed by means of the normal start sequence. Further operator intervention is required to bring the diesel up to rated speed.

3.3.1.13.3 Normal stop. Upon initiation of a stop sequence, the engine shall be brought to idle speed and maintained until proper cool-down, followed by the shut-down.

3.3.1.14 Governing system. The governing system shall be of the mechanical-hydraulic type or the electro-hydraulic type, as specified (see 6.2).

3.3.1.14.1 Mechanical-hydraulic type. If the mechanical-hydraulic type is used, speed characteristics as defined by MIL-STD-178 shall be as follows:

(a) Steady-state speed regulation limit: 1.0 percent.
(b) Maximum steady-state governing speed band: plus and minus 0.2 percent of rated speed.
(c) Minimum range of speed regulation changer: 0 to 5.0 percent of rated speed.
(d) Minimum range of speed changer: 90 to 105 percent of rated speed.
(e) Maximum permissible momentary underspeed and overspeed: 4.0 percent of rated speed for isochronous operation, 2.5 percent of rated speed for operation with a speed regulation of 3.0 percent.
(f) Maximum permissible recovery time in seconds following underspeed and overspeed: 2.
(g) Prescribed speedband for determination of recovery time: plus and minus 0.5 percent of rated speed.
(h) Range of speed regulation changer: 0 to 5.0 percent.

3.3.1.14.1.1 Speed changer for ac generator set. Unless otherwise specified (see 6.2), a motor-operated speed changer shall be provided on the generator set such that the frequency, synchronizing and load distribution of the connected ac generators are suitably controlled. Unless otherwise specified (see 6.2), speed changer motors shall conform to MIL-M-17059 and shall be of the following classification:

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Service - A.
Rating - 115 volts.
Revolutions per minute (r/min), horsepower (hp) and maximum torque as required by the synchronizing device.
Speed - varying.
Duty - intermittent, 1/2 hour.
Enclosure - totally enclosed.
Phases - single phase.
Bearings - ball or sleeve.
Ambient temperature - 50 degrees Celsius (°C).
Insulation - in accordance with MIL-E-917.
Mounting - horizontal.

3.3.1.14.2 Electro-hydraulic type. If an electro-hydraulic governing system is used, the system shall be speed and load sensing conforming to type DE of MIL-G-21410 and as specified in 3.3.1.14.2.1 through 3.3.1.14.2.7.

3.3.1.14.2.1 Maintenance of speed control. Under the abnormal conditions listed below, the system shall maintain speed control to the extent that the generator set shall not overspeed or shut down and the prime mover shall carry load and remain on the line. In addition, the specified performance shall automatically be resumed upon restoration of normal conditions.

(a) A three-phase or single-phase short circuit suddenly applied to the generator output terminals with the generator set delivering any load under normal operating conditions.
(b) Loss of generator excitation; that is, sudden removal of generator excitation with the set delivering any load under normal conditions. Overspeed is defined as 105 percent rated speed.

3.3.1.14.2.2 Hydraulic system. Hydraulic system shall be completely self-contained, separate from the prime mover lube oil system, and independent of any other ship system. The system shall consist of an electrical hydraulic transducer with gear driven pump, oil reservoir, accumulator (if needed), and any necessary filters. Piping shall be stainless steel or other corrosion-resistant material conforming to MIL-STD-777 (MIL-STD-438 for submarine application). The system shall stop the flow of fuel to the prime mover under the following conditions:

(a) Loss of control oil.
(b) Loss of signal or power supply to the governor electrical components.

For start-up purposes, a means of overriding the governor controls shall be provided. Corrodible internal cavities of the electrical hydraulic transducer shall be continuously flooded with oil.

3.3.1.14.2.3 Centrifugal backup governor. When specified (see 6.2), a centrifugal governor, which can be set at a predetermined speed level slightly higher than the electric governor so that it shall assume control of the prime mover upon failure of the electric governor, shall be furnished in conjunction with the electric governor. In such case, the shut-down provisions specified in 3.3.1.14.2.2 shall not apply.

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3.3.1.14.2.4 Permanent magnet alternator. A permanent magnet alternator conforming to MIL-G-3124 shall be furnished to supply power and speed signal to the governor. The alternator shall be short circuit stabilized.

3.3.1.14.2.5 Speed changer. A speed changer shall be provided and shall have a range of adjustment and fineness of control as specified in MIL-G-21410. The device may be single turn or multi-turn and shall be so connected as to raise speed by clockwise rotation. When a multi-turn device is used, a turn counting knob shall be provided. The device shall be motor operated and provide for both local and remote control. Electrical type (such as resistors in series) hi-speed and lo-speed stops shall also be provided with provision for adjustment up to 10 percent above (hi-speed stop) and 10 percent below (lo-speed stop) 60 hertz (Hz) speed. Stops shall be set at plus or minus 5 percent. Provision shall also be incorporated to locally override the hi-speed stop to permit gradual increase in speed up to overspeed under full control of the governor in order to check overspeed trip settings; also, to check low engine speed control of type IV sets. If a centrifugal governor is provided in conjunction with the electric governor (see 3.3.1.14.2.3), it may be used as the hi-speed stop provided that it can be readily adjusted manually to permit gradual increase in speed under its control up to the overspeed trip setting.

3.3.1.14.2.6 Provisions for kilowatt load division. Governors furnished with ship service and ship service/emergency sets shall be provided with both droop and differential real current compensation to provide for kilowatt load sharing. Means for selecting either mode of operation shall be provided. For emergency sets, the governors shall have provisions necessary for transferring power from or to other ship's generator sets, motor generator sets, or shore pow connections.

3.3.1.14.2.7 Governor system. Governor system shall be such that all electrical components can be permanently connected to the power and speed signal source and withstand any combination of frequency and voltage encountered during start-up, operation, or shut-down without damage.

3.3.1.15 Painting. Sets shall be painted in the same manner as for commercial delivery. Internal oil wetted surfaces shall not be painted. No paint, plastic or zinc coating shall be applied to the interior of gear housings, lubricating oil sumps, or any other interior surfaces that will be in contact with lubricating fuel or hydraulic oil.

3.3.1.16 Performance requirements

3.3.1.16.1 Load and overload. The generator set shall deliver rated load (rated kW at rated power factor (p.f.)) continuously, and 110 percent rated kW (rated p.f.) for 2 hours without damage to any part, in addition to generator overload criteria specified in MIL-G-3124.

3.3.1.16.2 Speed (r/min). The rated r/min of the diesel engine and the generator, when direct connected as specified in 3.3.1.5, shall conform to the requirements of the generator as specified herein.

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3.3.1.16.3 Protection.

3.3.1.16.3.1 Overspeed protection. As protection against dangerous engine overspeed due to failure of the regulating governor, its linkage or other engine malfunction, each engine shall be equipped with a separate and independent speed limiting device that shall prevent the engine from exceeding rated speed by more than 15 percent by cutting off completely the fuel and air to the engine. This device shall require manual resetting and shall not require electric power for proper operation.

3.3.1.16.3.2 Oil flow detection alarm. If the generator bearing is pressure-lubricated, an alarm system shall detect low oil flow at the generator bearing whenever the generator is rotating.

3.3.1.16.4 Vibration requirements. The generator set shall meet the vibration requirements in accordance with types I and II of MIL-STD-167-1. Each generator set shall meet the requirements of MIL-STD-167-2, type III, torsional vibration for both the normal operating condition and the single cylinder misfire conditions. Normal operation shall be defined as follows:

(a) No load: 0 to 115 percent rated speed continuous operation.
(b) Zero to rated load; 70 to 115 percent rated speed continuous operation.
(c) 115 to 180 percent rated load; 70 to 115 percent rated speed maximum two second operation.

When required, a forced response vibration analysis shall be performed in accordance with MIL-STD-167-2, type III, for each condition using harmonic excitation torques from the diesel engine and exciting torques from the generator (see 6.3).

3.3.1.16.5 Shock requirements. Unless otherwise specified (see 6.2), the diesel generator set and associated accessories shall meet the shock requirements specified in appendix A.

3.3.1.16.6 Short circuit requirements.

3.3.1.16.6.1 Type I sets. Generator sets shall withstand, without damage, a short circuit of the switchboard (assuming 25 feet of interconnecting cable from each generator power terminal to switchboard), with the quantity, type, and size of cables as specified (see 6.2).

3.3.1.16.6.2 Type II, III, and IV sets. Generators and associated excitation systems shall withstand without damage to any part three-phase and single-phase (line-to-line) short circuits at the generator terminals for a period of time as determined by $I^2t = 180$, where I is the sustained value of line current in per unit and t is the time in seconds. Unless otherwise specified (see 6.2), the sustained value of current during a three-phase or single-phase short circuit shall be not less than 3.2 times rated.

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malfunction indicator lamp to light when the frequency drops to 55 plus or minus

3.3.1.16.7 Parallel operation.

3.3.1.16.7.1 Ship service and ship service/emergency sets. Unless otherwise specified (see 6.2), ship service and ship service/emergency generator sets of the same size, type, and design on the same contract or order shall operate in parallel with each other throughout the permissible load ranges of the diesel engines. Generator sets of different sizes, types and design shall operate in parallel when specified (see 6.2).

3.3.1.16.7.2 Emergency generator sets. The emergency diesel generator sets shall be constructed for parallel operation and division of real and reactive power. The set, however, is not intended for continuous parallel operation but shall operate in parallel under manual or automatic control for short periods for the purpose of transferring real and reactive power from or to other ship's generator sets, motor generator sets, or shore power connections. The necessary governor and voltage regulator adjustments shall be provided and adjusted so that operation during such transfer of power shall be stable and controllable under any required operating condition.

3.3.1.16.8 Electromagnetic interference (EMI) control. Unless otherwise specified (see 6.2), EMI control shall be in accordance with part 5 or part 6, as applicable, of MIL-STD-461. Any power line filters or capacitors installed for the control or reduction of EMI shall conform to the requirements of MIL-STD-461 part 1 and approval for the use of such devices is required.

3.3.1.16.9 Power output ratings. Unless otherwise specified (see 6.2), generators shall have the following power output kW ratings based on 80 percent lagging power factor:

kW ratings	
30	1500
60	2000
100	2500
200	3000
300	3600
500	4000
750	5000
1000	8000

The kW rating shall be the net power for continuous use. If the generator has any auxiliary equipment such as a static excitation system that receives power from the generator and functions as an essential part of the basic generator set, the generator shall have capacity in excess of the net power rating to furnish this power. Such capacity shall be the gross kW rating.

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3.3.2 Type I - dc, single engine driven sets for surface ships.

3.3.2.1 Rating and characteristics.

kW rating	As specified (see 6.2).
Voltage	120 volts, two-wire or 120/240 volts, three-wire as specified (see 6.2).
R/min	720 to 1800, inclusive as specified (see 6.2).
Generator enclosure	Dripproof.
Ambient temperature	50°C.
Type of insulation	Class B, F, H or N in accordance with MIL-E-917 as specified (see 6.2).

3.3.2.2 Assembly. Each dc generator set shall consist of the following parts plus such other accessories as necessary to make the set a complete, self-contained unit when connected to the ship's fuel oil, electric, exhaust and water systems:

(a) Diesel engine prime mover.
(b) Diesel engine accessories.
(c) Dc generator.
(d) Balance coils (for three-wire generators).
(e) Generator field rheostat.
(f) Starting equipment.
(g) Governing system.

Unless otherwise specified (see 6.2), exciters, voltage regulators, field discharge resistors, battery charging generators, or starting batteries shall not be required.

3.3.2.3 List, pitch, roll, and trim. Type I sets shall operate in accordance with the performance requirements of this specification, shall maintain lubrication, and shall experience no loss of liquids under the following conditions:

(a) When the ship is permanently trimmed down by bow or stern as much as 5 degrees from the normal horizontal plane.
(b) When the ship is permanently listed up to 15 degrees to either side of the vertical.
(c) When the ship is pitching 10 degrees up or down from its normal horizontal plane.
(d) When the ship is rolling up to 45 degrees to either side of the vertical.

3.3.2.4 Combined unit voltage regulation. When driven by the intended diesel engine prime mover, the following requirements shall apply:

(a) With the field rheostat so adjusted that the generator is operating at rated voltage, current and r/min, the rise in voltage shall be not more than 8.0 percent when the load is reduced from rated load in 20-percent steps to zero.

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(b) With the field rheostat so adjusted that the generator is operati... at rated voltage, at no load, and at speed corresponding approximately to no load, the drop in voltage shall be not more than 12.5 percent when rated load is applied in 20-percent steps from zero load up to the rated output.

(c) At any point on the voltage-load curves obtained as specified in (a) and (b), there shall be no rise in voltage with increase in load, nor any drop in voltage with drop in load.

3.3.2.5 Combined unit parallel operation. Unless otherwise specified (see 6.2), the generators shall operate in parallel without voltage regulators or equalizers. At any given load division under the conditions of test specified in 4.6.15.1, the current load of any generator, expressed as a percentage of its rated full-load current, shall not differ from the total system current load expressed as a percentage of the total rated full-load currents of all connected generators by more than 7.5 percent for generators of the same kW rating, or 15.0 percent for generators of different kW ratings. (Example: If four 120/240 volt, dc generators, A, B, C, and D, rated 60, 100, 100, and 100 kW, respectively, are operating in parallel and at one point are delivering 165, 194, 195, and 196 amperes, respectively, with no current in the neutral lead, generator A would not be meeting specification requirements. They would all be meeting specification requirements at a certain point, however, if they were delivering 160, 196, 197, and 197 amperes, respectively, with no current in the neutral lead). Where there are currents flowing in the neutral leads of three-wire generators operating in parallel, the larger of the two main line currents in each generator shall be used to determine if parallel operation requirements are being met.

3.3.2.6 DC generators. Generators shall conform to MIL-G-3111.

3.3.2.7 Balance coils. Balance coils shall conform to MIL-T-15108, and shall be of the following classification:

Rating - as required by the dc generator.
Duty - continuous with 2-hour, 25-percent overload.
Mounting - as specified (see 6.2).

3.3.2.8 Generator field rheostats. Unless otherwise specified (see 6.2), generator field rheostats shall conform to MIL-R-15109 and shall be of the following classification:

Enclosure - dripproof.
Rating - as required by the generator.
Form - P.
Duty - continuous.
Mounting - switchboard.
Ambient temperature - 50°C.
Insulation - class C.

3.3.3 Type II - ac, single engine driven sets for surface ships.

3.3.3.1 Rating and characteristics. The rating and characteristics of the generator set shall be as follows:

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Rated capacity	as specified (see 6.2).
R/min	as specified (see 6.2).
Voltage	450 volts ac.
Frequency	60 Hz.
Phases	three.
Power factor	0.8 lagging.
Ambient temperature	50°C.
Type of insulation	class B, F, H or N of MIL-E-917 as specified (see 6.2).
Generator enclosure	500 kW and over - water-air-cooled under 500 kW - dripproof.
Voltage regulation	in accordance with type II or type III of MIL-R-2729.

3.3.3.2 Assembly. Each ac generator set shall consist of the following parts plus such other accessories as are necessary to make the generator set a complete, self-contained unit that functions without dependence on any other ship auxiliary when connected to the ship's fuel oil, electric, exhaust and water systems:

(a) Diesel engine prime mover.
(b) Diesel engine accessories.
(c) AC generator.
(d) Excitation and voltage regulation system.
(e) Starting equipment.
(f) Governing system.

Unless otherwise specified (see 6.2), field discharge resistors, starting batteries, or switches for synchronizing motors shall not be required.

3.3.3.3 List, pitch, roll, and trim. Type II set shall operate in accordance with the performance requirements of this specification, shall maintain lubrication, and shall experience no loss of liquids under the following conditions:

(a) When the ship is permanently trimmed down by bow or stern as much as 5 degrees from the normal horizontal plane.
(b) When the ship is permanently listed up to 15 degrees to either side of the vertical.
(c) When the ship is pitching 10 degrees up or down from its normal horizontal plane.
(d) When the ship is rolling up to 45 degrees to either side of the vertical.

3.3.3.4 Combined unit parallel operation. The ship service and ship service/emergency generators shall operate in parallel in accordance with the kilovoltamp (kVA) load (reactive power) division and circulating current requirements specified in MIL-R-2729 (see 4.6.15.2) and in accordance with the following kW load (active power) division requirements: At any specified load division (see 4.6.15), the actual kW load of any generator, expressed as a percentage of its continuous kW rating, shall not differ from the total system kW load, expressed as a percentage of the total continuous kW rating of all connected generators, by more than the following:

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(a) 5.0 percent for generators of the same size, type, design, and rating.
(b) 10.0 percent for generators of unequal size, type, design, and rating.

The current pulsation shall not exceed 2.0 percent of the rated current of one generator (see 4.6.15.2.1.1 and 4.6.15.2.2.1).

3.3.3.5 AC generators. Generators shall conform to MIL-G-3124, except as otherwise specified herein.

3.3.3.5.1 Location of terminals. The location and arrangement of the generator main terminals, cooler connections, and static exciter (if used) terminals shall be as required by the installation. A mounting pad or flange shall be provided for the attachment of a shipbuilder furnished terminal box for the main terminals and for the exciter terminals if required.

3.3.3.5.2 Generator constants. Any variation between the actual and calculated generator reactances values of more than 10 percent shall be referred to the Command or agency concerned for approval. The short circuit ratio shall be not less than 0.8. Generators shall be electrically stable under all operating conditions.

3.3.3.6 Voltage regulation and excitation system. The voltage regulation and excitation system shall conform to type II or III of MIL-R-2729. If type III is used, the power rectifiers shall be mounted in the generator cooling air stream and shall be readily accessible for examination, removal, disassembly, or replacement. Reactors, load current transformers, and other excitation components shall be mounted in a drippproof enclosure arranged for deck mounting. Voltage regulator components, controls, and switches shall be arranged for mounting in accordance with MIL-S-16036.

3.3.3.6.1 Differential reactive current compensation. Differential reactive current compensation shall also be provided in addition to individual reactive current compensation in order to provide automatic kilovar load sharing. Means for selecting either mode of operation shall be provided.

3.3.4 Type III - ac, twin engine driven sets for surface ships. The requirements for type III sets shall be as specified in 3.3.4.1 through 3.3.4.5 and as specified for type II sets in 3.3.3.

3.3.4.1 Generator sets. Type III generator sets shall be arranged with the engines in-line and directly connected to the generator through manually-operated clutches. The engines and generator shall be mounted on a common sub-base. Accessories as necessary to make the set a complete, self-contained unit that functions without dependence on any other ship's auxiliary when connected to the ship's fuel oil, electric, air, exhaust, and water systems shall be provided. The set shall operate in accordance with the performance requirements specified herein at full load continuously with both engines running or at 50 percent load continuously with either engine running and the other declutched and secured.

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3.3.4.2 Contacts. Mechanically operated contacts shall be included in the clutches and starting system which shall close when the clutches are engaged and the unit is set for starting both engines. These contacts are intended to be used in an indicating lamp circuit to be supplied by the shipbuilder when the unit is to be installed for automatic start-up under emergency conditions. Contacts shall be suitable for a 450-volt, single-phase, 2-ampere ac system.

3.3.4.3 Pumps. Engine driven pumps shall be provided for supplying oil to the generator bearing. The pumps shall be arranged so that oil is delivered to the bearing with either of the engines declutched and secured. Return oil shall drain to a separate sump tank installed on the sub-base near the generator or the engine oil sump. A scavenging pump shall be furnished if required. The generator bearing lube oil system shall be furnished complete with pumps, sump, and piping. If a separate sump tank is used, it shall incorporate an oil level indicator.

3.3.4.4 Starting system. Each engine shall be provided with its own air or electric starting motor. Provision shall also be made for manually selecting either single engine starting or simultaneous twin engine starting.

3.3.4.5 Division of load. At full load, the engines shall divide load so that the difference in load carried by each engine shall not exceed 2.5 percent of the generator kW rating.

3.3.5 Type IV - ac, single engine driven sets for submarines.

3.3.5.1 Assembly. The diesel generator set shall consist of the diesel engine, generator, and such attached and unattached accessories as to make a complete generator set that functions in accordance with this specification without dependence on any other ship's auxiliary when connected to the ship's fuel, exhaust, electric and water system. Generator set hold down bolt holes provided for securing the equipment to the ship's foundation shall use non-fitted bolts on one side and fitted bolts on the other in order to ensure there are no adverse effects on operation or alignments as a result of hull deflections.

3.3.5.2 Rating and characteristics. The rating and characteristics of the generator set shall be as follows:

Rated capacity	kW ratings for surface and snorkel shall be as specified (see 6.2).
R/min	as specified (see 6.2).
Voltage	450 volts ac.
Frequency	60 Hz.
Phases	three.
P.f	0.8 lagging.
Ambient temperature	50°C.
Type of insulation	class B, F, H or N of MIL-E-917 as specified (see 6.2).
Generator enclosure	water-air-cooled.
Starting	air.
Voltage regulation and excitation system	in accordance with type II or type III of MIL-R-2729.

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3.3.5.2.1 Continuous operation under specified loads. The type IV diesel generator set shall continuously operate at the rated snorkel and surface loads as specified (see 6.2) under conditions of inlet vacuum and exhaust back pressure as specified (see 6.2). The generator set shall also be as specified below:

(a) Under snorkel and surface conditions with a base load on the set as specified (see 6.2), generator set shall start a hydraulic pump motor of horsepower rating, inrush current and p.f. as specified (see 6.2), and maintain generator terminal voltage within plus or minus 18 percent of initial terminal voltage. The terminal voltage shall return to and remain within plus or minus 3 percent of the final sustained voltage within 2 seconds of the initiation of the start.

(b) Generator set shall start a motor generator set of kW rating and inrush current as specified (see 6.2) with no other load on the set.

3.3.5.3 List, pitch, roll and trim. Unless otherwise specified (see 6.2), the diesel generator set and associated auxiliaries shall operate in accordance with the performance requirements of this specification, and avoid loss of oil under the most unfavorable combination of the following conditions of trim, pitch, list, and roll:

Trim	- plus or minus 7 degrees.
Pitching	- plus or minus 10 degrees.
Time of complete pitch cycle	- 5 to 10 seconds.
List	- plus or minus 15 degrees.
Rolling	- 30 to 60 degrees.
Time of complete roll cycle	- 8 to 15 seconds.

The set shall not leak oil under trim angles up to 30 degrees. This is not an operating condition. Degrees indicated are measured from the vertical to either side for list and roll, and from the horizontal fore and aft plane, up or down, for trim and pitch. The conditions of permanent list and roll or trim and pitch shall not be considered additive. The resultant angle of inclination from the following combinations shall be met.

(a) List and trim.
(b) List and pitch.
(c) Roll and trim.
(d) Roll and pitch.

3.3.5.4 Switches. The contractor shall provide the following items:

(a) A low engine speed shutdown control for snorkel operation which shall include a shutdown device, switch, and governor solenoid operating from a 115-volt, 60-Hz, ac source. The device shall be adjustable between approximately 50 to 90 percent of rated speed, and shall actuate the switch when the speed falls to a minimum of 80 percent of rated speed. The switch shall de-energize the governor solenoid which in turn shall shut the engine down. The governor solenoid shall be provided with a manual override for

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emergency stops and for set starting with the solenoid de-energized. The low speed switch shall be of the double pole, single throw type, 3 to 5 pounds per square inch (lb/in^2) range, set to open on decrease of lube oil pressure. An identical low speed switch shall trip the generator circuit breaker when the speed falls to a minimum of 80 percent of rated speed. The settings of these switches shall be such that the generator breaker switch operates first.

(b) An overspeed trip actuated switch for tripping the generator circuit breaker.

(c) A 1/2-inch connection in the blower inlet and discharge for installation of a differential temperature thermometer.

(d) A 1/2-inch connection in the engine crankcase for manometer connection.

(e) A motor-operated speed changer, single phase, 115-volt ac, readily reversible, for speed sensing type governors.

3.3.5.5 Generators. The generator with kW rating as specified (see 6.2), shall conform to MIL-G-3124, except as otherwise specified herein.

3.3.5.5.1 Construction. Reduced weight and size shall be major factors in the generator construction. The generator shall be of single, sleeve type bearing and shall have the engine end of the shaft piloted in the engine crankshaft. The generator half of the coupling flanges shall be an integral part of the generator shaft forging. A sight flow fitting shall be provided at the bearing oil discharge.

3.3.5.5.2 Enclosure. The lower half (up to the bottom of the shaft) of the generator enclosure shall be watertight, and the upper half of the enclosure shall be spraytight.

3.3.5.5.3 Air cooler. The generator air cooler shall be in accordance with class 1 of MIL-C-19836 and, if required, shall be constructed for the submergence design pressure and submergence test pressure specified (see 6.2). Cooling water shall be furnished from the engine seawater system.

3.3.5.5.4 Thermometers. Requirements covering bearing thermostatic switch and temperature indicating equipment in accordance with MIL-G-3124 and air temperature alarm in accordance with MIL-C-19836 shall not apply. In lieu thereof, continuous reading resistance type thermometers shall be provided for measuring the temperature of the air in and out of the cooler and bearing, and shall be mounted on the engine gauge board.

3.3.5.5.5 Terminals. The location and arrangement of the generator main terminals, cooler connections, field rectifier terminals (if rectifiers are mounted in air stream and terminals are located outside generator enclosure), and heater connections shall be as required by the installation. A mounting pad or flange shall be provided for the attachment of a shipbuilder furnished terminal box for the main terminals and rectifier terminals if required.

3.3.5.6 Voltage regulation and excitation system. The voltage regulation and excitation system shall be type II or III in accordance with MIL-R-2729 as specified (see 6.2).

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3.3.5.6.1 Voltage regulator. The voltage regulator shall be of the following classification:

(a) Enclosure..... protected for switchboard mounting.
(b) Regulators per switchboard... one.
(c) Operation..... parallel (see 3.3.1.16.7).

3.3.5.6.2 Mounting. Voltage regulator components and parts, except the manual control unit, the voltage adjustment unit, the control switch, field flashing switch, and droop compensation transformers shall be mounted in one box. The control units and switches shall be constructed for back of board mounting, front of board operation.

3.3.5.6.3 Power rectifiers. If type III voltage regulation and excitation system is used, the power rectifiers shall be mounted in the generator cooling air stream and shall be readily accessible for examination and replacement. Unless otherwise specified (see 6.2), other components of static exciter shall be suitable for mounting within the switchboard enclosure.

3.3.5.6.4 Field flashing. An external source of dc power shall be available and may be used for field flashing of the ac generator. If field flashing is provided from an ac source through rectifiers, the flashing switch shall open both the ac and dc sides of the circuit in the "off" position.

3.3.6 Identification plates. The identification plates shall be attached to the part of the equipment which is not ordinarily renewed during its normal service life. These plates shall be located in a readily accessible position where they can be read at all times without danger to personnel. Identification plates shall be either type A, B or C in accordance with MIL-P-15024 and MIL-P-15024/5. Engraved, stamped or direct etched markings shall be filled with black paint, enamel, or lacquer. The information marked on the plates shall include the following items:

(a) Manufacturer's name, identification symbols, serial number, contract or order number, and date of manufacture.
(b) Salient design characteristics, for example, type, frequency, voltage, and capacity.
(c) National stock number, if available.

3.3.7 Airborne and structureborne noise.

3.3.7.1 Airborne noise. Unless otherwise specified (see 6.2), the sound pressure level of the diesel generator set at rated load, including all attached and detached accessories shall not exceed the following levels (see 4.6.23.1).

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Octave band center frequency, Hz	Sound pressure level, dB re 20 μPa
31.5	91
63	88
125	85
250	82
500	79
1000	76
2000	73
4000	70
8000	67

3.3.7.2 Structureborne noise. The structureborne noise levels will be specified at any measurement location and in any direction shall not exceed the levels specified (see 6.2 and 4.6.23.2). The structureborne noise levels of the generator set with all attached accessories, and the levels of detached accessories shall be measured at rated load. Details on whether generator sets and detached units must be resiliently mounted and the type of resilient mount to be used shall be as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. The first generator of a design, type, and size offered for delivery shall be subjected to the examination and tests specified in table I (see 6.3). First article inspection shall be required after any change in design or use with a new generator design which affects the performance characteristics.

TABLE I. First article and quality conformance inspection.

Description of tests	Requirement	Test method	First article inspection	Quality conformance inspection
Examination	3.3.1	4.5	X	X
Generator air gap	3.3.7	4.6.1	X	X
Five- (or eight-) hour continuous test at full load	3.3.1.16	4.6.3	X	X
Two-hour, 110 percent rated load test	3.3.1.16.1	4.6.4	X	X
Governor	3.3.1.14	4.6.5	X	X
Voltage regulation with regulator in field circuit (ac generator sets only)	3.3.3.1	4.6.6	X	X
Voltage regulation of dc generator sets	3.3.2.4	4.6.7	X	X
Commutation	3.3.2.6	4.6.8	X	X
Effectiveness of enclosure Type I sets Type II and III sets Type IV sets	 3.3.2.1 3.3.3.1 3.3.5.5.2	4.6.9	X	
Internally excited vibration (mechanical balance)	3.3.1.16.4	4.6.10.1	X	X
Torsiograph	3.3.1.16.4	4.6.10.2	X	
Lateral vibration	3.3.1.16.4	4.6.10.3	X	
Lubrication	3.3.1.7	4.6.11	X	
Overspeed trip	3.3.1.16.3.1	4.6.12	X	X
Oil flow detection	3.3.1.16.3.2	4.6.13	X	X

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TABLE I. First article and quality conformance inspection - Continued.

Description of tests	Requirement	Test method	First article inspection	Quality conformance inspection
Starting	3.3.1.13	4.6.14	X	4.6.14.5 and 4.6.14.6 only
Parallel operation		4.6.15	X	
Type I sets	3.3.2.5			
Types II, III and IV sets	3.3.3.4			
Division of load type III sets	3.3.4.5			
Inclined operation		4.6.16	X	
Type I sets	3.3.2.3			
Types II and III sets	3.3.3.3			
Type IV sets	3.3.5.3			
Weight and center of gravity	3.3.1	4.6.17	X	
EMI control	3.3.1.16.8	4.6.18	X	
High impact shock	3.3.1.16.5	4.6.19	X	
Reliability demonstration	3.3.1.2	4.6.20	X	
Maintainability demonstration	3.3.1.2.5	4.6.21	X	
Snorkel operation (type IV)	3.3.5.2.1	4.6.22	X	
Airborne and structureborne noise	3.3.7	4.6.23	X	

4.4 Quality conformance inspection. Quality conformance inspection shall consist of tests specified in table I (see 6.3). If the test data reveal variations beyond a normal manufacturing tolerance, the government inspector at his discretion, may order more tests to demonstrate that the generator set meets the requirements of this specification.

4.5 Examination. The generator set shall be examined for access to accessories, such as oil sight flow gauges, lube oil fill and drain, air and oil filter, instruments, and so forth. Each generator set shall be thoroughly examined for proper alignment and to ascertain that the material, workmanship, and construction conform to this specification. The fit of parts shall be observed with particular reference to the interchangeability of such parts as are likely to require replacement during the normal service life of the generator set.

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4.6 Test methods.

4.6.1 Generator air gap. The minimum air gap between the rotor and stator iron shall be carefully measured. The air gap shall be measured by steel feelers or gauges. The measurements shall be made in at least eight places approximately 45 degrees apart, with at least one of the measurements made at the bottom. Each measurement shall be within 10 percent of the design air gap indicated on the contractor's drawing.

4.6.2 Preliminary run. The generator set shall be operated at full load for 45 minutes (2 hours for type IV generator set) to bring it up to or near its operating temperature. During this period the general operation of the set shall be observed and any necessary adjustments shall be made.

4.6.3 Five- (or eight-) hour continuous test at full load.

4.6.3.1 Five-hour continuous test at full load (for types I, II and IV generator sets). Immediately following the preliminary run, the generator set shall be operated at full load for 5 hours. The following information shall be taken at 30-minute intervals during the test:

(a) DC and ac generator sets:

(1) Time
(2) Ambient temperature.
(3) Fuel consumption.
(4) Engine fuel oil pressure.
(5) Engine lubricating oil pressure.
(6) Engine lubricating oil temperature.
(7) Water temperature.
(8) Speed of engine.
(9) Generator bearing temperatures.

(b) DC generator sets only:

(1) Volts, line to line.
(2) Volts, line to neutral (for three-wire sets).
(3) Current, line.
(4) Current, neutral (for three-wire sets).
(5) Generator frame temperature.
(6) Generator exhaust temperature.

(c) AC generator sets only:

(1) Volts, line-to-line.
(2) Current, line.
(3) Generator frame temperature.
(4) Generator armature temperature.
(5) Generator field current and voltage.
(6) Generator inlet and exhaust air temperature.
(7) Frequency.
(8) kVA and kW.
(9) Power factor.

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At the conclusion of the 5-hour period, temperatures of the rotating components of the generator shall be measured and shall be recorded immediately upon shutdown.

4.6.3.2 Eight-hour continuous test at full load (for type III generator set only). The generator set shall meet the requirements of the above 5-hour test except the test shall be run for 8 hours instead of 5 hours. Each engine connected to the generator with the other engine declutched shall be tested for 8 hours at 50 percent of the generator set full load rating. Sufficient information shall be determined to demonstrate that engines divide load within 2-1/2 percent of their individual rating.

4.6.4 Two-hour, 110 percent rated load test. Following any full load run, 110 percent rated load shall be applied for 2 hours. The operation of protective devices shall be checked during this test.

4.6.5 Governor tests. If a speed and load sensing governor is used, tests shall be conducted in accordance with MIL-G-21410. If a speed sensing governor is used, tests shall be conducted as specified in 4.6.5.1 through 4.6.5.5.

4.6.5.1 Load speed test. The load speed test, with generator excitation manually controlled for type I and automatically controlled for types II, III and IV, shall be performed as follows:

(a) At no-load, and with speed regulation set to zero, the generator set shall be adjusted to rated speed and voltage.
(b) The load shall be changed in 10.0 percent increments from no-load to 100 percent load and back to no-load. At each point, the voltage shall be adjusted to rated value, and the speed recorded.
(c) The load speed curve shall be plotted.

4.6.5.2 Sudden load change. The sudden load change test, with generator excitation manually controlled for type I and automatically controlled for types II, III and IV, shall be conducted as follows:

(a) With speed regulation set to zero the generator set shall be adjusted to rated speed, voltage, and load. Rated load shall then be removed.
(b) The generator set shall be operated at rated speed and 105 percent rated voltage. A 100 percent rated resistance load shall then be suddenly applied.
(c) Readings of overspeed, underspeed and recovery time shall be taken by means of an oscillograph or similar recording device. Sudden removal or application of load shall be accomplished by means of a circuit breaker installed between the load and the generator output terminals.

4.6.5.3 Steady state governing speedband. At 0, 25, 50, 75 and 100 percent rated load, the steady state governing speedband shall be recorded.

4.6.5.4 Range of speed changer. The range of speed changer test, with generator excitation manually controlled for type I and automatically controlled for types II, III and IV, shall be conducted as follows:

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(a) With speed regulation set to zero the generator set shall be adjusted to rated speed and voltage at no load. Operate the speed changer so that the speed is at the minimum specified value. Record speed. Operate the speed changer so that the speed is at the maximum specified value. Record speed.

(b) The generator set shall be adjusted to rated speed and voltage at 100 percent load. Operate the speed changer so that the speed is at the minimum specified value. Record speed. Operate the speed changer so that the speed is at the maximum specified value. Record speed and rate of change. The load and voltage may be maintained at rated values when measuring maximum speed.

4.6.5.5 Range of speed regulation changer. The diesel generator shall be adjusted to rated speed and voltage at no load. The generator set shall then be loaded to rated kW load to determine conformance to the requirement for the range of the speed regulation changer.

4.6.6 Voltage regulation test with regulator in field circuit (ac generator sets only). The requirements of MIL-G-3124 and MIL-R-2729 shall be met.

4.6.7 Voltage regulation of type I (dc generator sets). The voltage regulation test shall be conducted to determine conformance to 3.3.2.4.

4.6.8 Commutation type I (dc generator sets). During all tests, the commutation shall be noted at each load point. Particularly, it shall be noted if any injurious sparking occurs when the entire load is dropped or applied in one step.

4.6.9 Effectiveness of enclosure. Enclosure tests shall be conducted in accordance with MIL-E-2036 on the electrical portion of the generator set, when it is evident that the effectiveness of any enclosure has been altered by assembly as a part of the generator set.

4.6.10 Vibration.

4.6.10.1 Internally-excited vibration (mechanical balance). The mechanical balance shall be observed during the progress of all tests. Any unusual noises, excessive vibration, or other indication of malfunction shall be cause for rejection of the unit. Vibration readings in mils of double amplitude shall be taken on the engine and generator bearing brackets or shafts.

4.6.10.2 Torsiograph test. Generator set shall be tested in accordance with type III of MIL-STD-167-2.

4.6.10.3 Lateral vibration. A complete mathematical analysis of the lateral (whirl) vibratory characteristics of the rotating diesel generation system shall be made. This analysis shall clearly demonstrate that the system is free from any lateral critical frequency below 115 percent of the maximum rated speed, unless otherwise approved by the Command or agency.

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4.6.11 Lubrication. Performance of the lubricating system shall be determined by a careful observation and general examination of the parts. Limitation of the generator bearing temperatures, as specified in MIL-G-3124 or MIL-G-3111, as applicable, shall not be exceeded. This test may be made at no-load provided the bearing load is not influenced appreciably by the load condition of the generator.

4.6.12 Overspeed trip. The engine speed shall be gradually increased, under control of the governor, above rated speed until the overspeed trip functions. The r/min at which the trip functions shall be recorded.

4.6.13 Oil flow detection. Oil flow to the generator bearing with the engine not running shall be established to evaluate operation of the oil flow detection alarm system.

4.6.14 Starting tests. The tests specified in 4.6.14.1 through 4.6.14.4 shall be conducted on the first unit only. Tests specified in 4.6.14.5 and 4.6.14.6 shall be conducted on each unit.

4.6.14.1 Cold start. Before the test, the diesel generator set shall be shut down for at least 12 hours. Cold water at a temperature not greater than 35 degrees Fahrenheit (°F) shall be circulated through the engines for not less than 1 hour, or until all parts of the engine are cooled to a reasonably uniform temperature. The engine shall fire, continue operating, and provide rated voltage at the generator terminals within 10 seconds after the starting mechanism is set in operation.

4.6.14.2 Normal starting. Immediately following the cold start test (see 4.6.14.1), nine consecutive starts shall be made under normal ambient conditions. The average time required to start the unit, and the minimum starting r/min shall be determined. The engine shall fire and continue operating within 5 seconds after the starting mechanism is engaged.

4.6.14.3 Air starting. The following information shall be obtained if air starting is used:

(a) The minimum quantity of air expressed in cubic feet of free air at a gauge pressure of 14.7 lb/in^2 and 68°F for the cold start.
(b) The minimum quantity of air expressed in cubic feet of free air at a gauge pressure of 14.7 lb/in^2 and 68°F necessary for ten successive starts, including the cold start.
(c) The minimum air pressure at which the unit will start within the 5-second time limit

4.6.14.4 Electric starting. When electric starting is used, the starting motor current and voltage shall be recorded for each start.

4.6.14.5 Starting sequence. Tests shall be conducted to demonstrate that the start sequence conforms to the ship service emergency and ship service/emergency requirement, as applicable (see 3.3.1.13.2).

4.6.14.6 Stop sequence. Tests shall be conducted to demonstrate that the stop sequence requirements are met (see 3.3.1.13.3).

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4.6.15 Parallel operation test. This test shall be conducted on the generator set with an identical generator set.

4.6.15.1 Type I. With each generator operating at rated voltage and at a speed corresponding to approximately 20 percent load, the engine governors shall be adjusted for equal percentage division of load equal to approximately 20 percent of the combined kW rating of the paralleled generators. No further adjustment of the governor and rheostats shall be made for the duration of the test. The load shall be varied from 20 to 100 percent of the combined kW rating of the two paralleled generators in four approximately equal steps and back to 20 percent load in the same manner. The speed, line voltage, line current, kW load, field voltage and field current of each generator at each load point shall be recorded.

4.6.15.2 Types II and III (except emergency service). Parallel operation tests on ship service and ship service/emergency service generator sets shall be performed as specified in 4.6.15.2.1 through 4.6.15.2.2.

4.6.15.2.1 Electro-hydraulic governors. Where electro-hydraulic governors are required, parallel operation tests shall be performed in accordance with MIL-G-21410.

4.6.15.2.2 Mechanical-hydraulic governors. Where mechanical-hydraulic governors are required, parallel operation tests shall be performed as follows:

(a) This test shall be made with each generator connected to its intended diesel engine prime mover and with the automatic voltag regulator and exciter which are intended to be furnished with the generator set. Before the test, each governor shall have real load droop compensation set for equal real load division and each voltage regulator shall have reactive droop compensation set for equal reactive load division. With each generator at rated voltage and at a speed corresponding to approximately 20 percent load, the governors and voltage regulators shall be adjusted for equal percentage division of load at rated p.f. No further adjustment of the governor, voltage regulator and rheostats shall be made for the duration of the test. The load shall be varied from 20 percent to 100 percent of the combined continuous kVA rating of the parallel generators at approximately rated p.f. in four approximately equal steps and back to 20 percent load in the same manner. The speed, voltage, current, kW load, reactive kVA load, field voltage and field current of each generator at each load point shall be recorded.

(b) The current pulsation of paralleled generators shall be measured at 20, 40, 60, 80, and 100 percent of rated continuous load at 0.80 p.f. The current pulsation is defined as the differences between the maximum root mean square (rms) and minimum rms current values that occur during a cycle of pulsation, expressed as a percentage of rated full load rms current. The maximum and minimum values shall be determined by the pencil ammeter method using a portable ammeter conforming to MIL-M-16034. To obtain the current pulsation by this method, a generator shall be operated in parallel with another generator under conditions of steady load.

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The pointer will be observed to oscillate through a small angle. Due to the damping of the meter, the pointer will indicate less than the time pulsation in current. The glass cover over the scale shall be removed and a straightedge placed against the pointer. The pointer shall be moved slowly by the straightedge toward the maximum point of its swing until the pointer just perceptibly moves away from the straightedge. The pointer then reads the maximum value of rms current during a cycle of the pulsation. The minimum value of rms current shall be read similarly by placing the straightedge on the opposite side of the pointer and moving it toward the minimum point of the swing. The percent pulsation shall be

$$\frac{(I_{max} - I_{min})}{I_{rated}} \times 100$$

4.6.15.3 Type IV and emergency service (types II and III). Parallel operation tests shall be conducted to the extent necessary to demonstrate stable transfer of active and reactive power from one set to the other and with the voltage plus or minus 5 volts of rated.

4.6.16 Inclined operation. The completely assembled diesel generator set and attached accessories shall be operated at rated speed, with or without load, for at least 30 minutes in each of the following positions:

(a) Shaft inclined down 5 degrees (7 degrees for submarine application).
(b) Shaft inclined up 5 degrees (7 degrees for submarine application).
(c) Shaft horizontal, base inclined to left 15 degrees.
(d) Shaft horizontal, base inclined to right 15 degrees.

During these tests, it shall be ascertained that mechanical balance and vibration are satisfactory, that there is no pounding or grinding of bearings, and there is no leakage of oil from bearings or of cooling water.

4.6.17 Weight and center of gravity. The generator set and all detached accessories shall be weighed dry. The center of gravity of the generator set shall be determined by actual measurement, if feasible. Otherwise it shall be calculated using weights and center of gravity figures for components obtained by actual measurement.

4.6.18 EMI testing. Unless otherwise specified (see 6.2), the complete diesel generator set shall meet the EMI control requirements specified in MIL-STD-461 part 5 or part 6, as applicable, when tested in accordance with the procedures contained in MIL-STD-462.

4.6.19 High impact shock. Unless otherwise specified (see 6.2), the complete diesel generator set shall be tested in accordance with the requirements of appendix A to this specification (see 6.3 and appendix A).

4.6.20 Reliability demonstration test. The reliability demonstration test, if required, shall be as specified (see 6.2).

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4.6.21 Maintainability demonstration test. A maintainability demonstration test shall be in accordance with MIL-STD-471 except as specified (see 6.3).

4.6.21.1 Test conditions. The maintainability test shall be performed with the equipment or system installed in a manner which simulates an actual ship-board installation and the access envelope (see 3.3.1.2.7) shall be simulated with panels and screens. Test equipment tools, repair parts and maintenance literature, at least in the form of preliminary manuals covering maintenance instruction, shall be made available for the tests. Test team, facilities, and support material shall represent the normal shipboard resources. Test mechanics shall be given no outside assistance unless approved by the acquisition activity. The following shall be recorded, monitored, and certified by the acquisition activity's representative:

(a) Data collected, including man-hours, clock-hours, maximum number of active mechanics, part identification, and documentation of the specified maintenance tasks.
(b) Factors which influenced the data.
(c) Identification of other parts removed or moved to permit replacement of the part for which a replacement time has to be demonstrated.
(d) Computation or measurement of the required replacement times.
(e) Deficiencies.
(f) Recommendations.
(g) Results of retest, if applicable.
(h) Certification by an acquisition activity's representative of the data obtained and whether or not the contractor task time requirements have been met.

4.6.22 Snorkel operational test.

4.6.22.1 Temperatures. The supplier shall establish maximum limits for exhaust gas temperatures and blower intake and discharge temperature differential. The engine shall then be operated at various snorkel loads and rated r/min at various blower intake vacua and exhaust back pressures in order to determine the engine output limits at which the engine will not exceed exhaust temperature or blower differential temperature limits as established above.

4.6.22.2 Snorkel operation. The engine shall be operated continuously for 160 hours rated snorkel load and r/min. The exhaust back pressure at the manifold outlet shall be 6 lb in^2. Intake air pressure shall be 2 inches of mercury (Hg), vacuum. Fuel consumption shall be determined and submitted to the command or agency concerned.

4.6.22.3 Simulating loss of depth control. The engine shall be operated at the greatest load which will not exceed temperature limits established in 4.6.22.1. Automatic control of the blower intake and exhaust valves shall be arranged to gradually decrease the blower intake pressure from 2 inches of Hg below atmospheric pressure up to 7 inches of Hg below atmospheric pressure, and at the same time increase exhaust back pressure from 6 lb/in^2 up to 12 lb/in^2. The time required to reach these maximum pressure conditions shall be 3.5 minutes. The control valves shall then permit the engine to return to snorkel operations

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1 minute. The engine shall operate 5 minutes under snorkel conditions. The entire cycle, therefore, will require 9.5 minutes. The unit shall operate on this cycle 500 times to demonstrate the mechanical reliability of the engine.

4.6.22.4 Simulated wave action. The air inlet pressure shall be constant throughout this test and shall be 2 inches of Hg below atmospheric pressure. The exhaust back pressure shall be regulated for a standard pressure of 6 lb/in^2 with the damper valve in exhaust line controllable to raise the pressure to 8 lb/in^2. The cycle of operation shall be for the pressure to rise to its maximum in a period of 25 seconds, returning to the standard pressure in 5 seconds and a rest period of 30 seconds or a total cycle time of 60 seconds. This cycle of operation shall be repeated for a period of 48 hours at rated snorkel load and r/min.

4.6.22.5 Simulated submerged starting test to remove water from the snorkel tube. The engine shall be started with the exhaust line closed until the back pressure reaches a maximum of 20 lb/in^2 in approximately 6 seconds. The exhaust line shall then be opened to reduce the back pressure to the normal pressure of 6 lb/in^2. The engine shall then be shut down, and the test shall be repeated. A total of 250 starts shall be made. One of the starts shall be made from the cold ambient condition after the engine has been shut down for at least 24 hours.

4.6.23 Airborne and structureborne noise tests.

4.6.23.1 Airborne noise. The test shall be performed on the completely assembled generator set in accordance with MIL-STD-740-1.

4.6.23.2 Structureborne noise. The structureborne noise testing shall be performed in accordance with MIL-STD-740-2. The structureborne noise levels shall be measured in the vertical direction and in two other mutually perpendicular directions on each "foot" or attachment point of the diesel generator unit and the detached accessories in order to determine conformance to 3.3.7.2. Structureborne noise measurements shall also be taken at the following three locations to aid in future maintenance and diagnosis:

(a) On each end of the unit, near its axis, in a radial direction.
(b) On one end of the unit, near its axis, in an axial direction.

4.7 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein (see 6.3).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.11.)

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5.1 General.

5.1.1 Navy fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

5.2 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Generators and voltage regulators. Generators and voltage regulators shall be cleaned and preserved level A in accordance with MIL-E-16298, MIL-R-2729, and MIL-E-17555, as applicable.

5.2.1.2 Diesel engine. The basic diesel engine shall be cleaned and preserved in accordance with type IV, method II, of MIL-E-10062 to the extent specified. Manifolds, pumps, turbochargers and other attached accessories shall not be removed from the engine during cleaning and preservation unless the removal is considered essential for complete coverage of the preservative.

5.2.1.2.1 Lubricating oil, fresh and sea water systems. Preservative of type as applicable, grade 30, in accordance with MIL-L-21260, shall be used throughout the engine, fresh water, sea water, and lubricating oil systems in lieu of the P-2, P-5, and P-9 preservative specified in MIL-E-10062. When required, contractors may impose a maximum zinc content limit. Clean filter elements shall be installed in the fuel and lubricating oil systems after preservation is completed. In addition, the fresh water system with thermostats removed shall be drained and completely dried with warm air prior to preservation. If it is impractical to perform a drying-out procedure, the cooling system shall be flushed with a soluble oil conforming to MIL-I-24453, which will emulsify the water and remove it on draining. The system shall then be flushed with the specified preservative and the thermostats replaced. Preservative oil shall be drained from all systems including the oil pan.

5.2.1.2.2 Fuel oil system. Preservative in accordance with MIL-L-21260, type as applicable, grade 10W, shall be used throughout the fuel system.

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5.2.1.3 Packaging. The diesel generating set with attached accessories shall be individually unit protected in accordance with MIL-P-116, method II. To ensure optimum desiccation, engine inlets and outlets shall not be sealed. One of the following two packaging methods shall be used as specified (see 6.2):

(a) Method II(a) - nonreusable floating bag. The flexible bag shall conform to MIL-P-116 and MIL-B-131, class 1. Humidity indicators shall conform to MIL-I-26860, (plug type). Projections and sharp corners of the engine unit shall be cushioned as specified in 5.3.3. Whenever hold-down bolts go through the barrier bag, gaskets fabricated of material conforming to F112400E45M6 of MIL-G-12803 shall be secured to both the inside and outside of the bag with adhesive conforming to MMM-A-260 for fiber gaskets and MMM-A-1617 for rubber gaskets. Desiccant shall be uniformly distributed throughout the package to obtain optimum effectiveness.

(b) Method II(b) - reusable floating bag. A reusable container conforming to MIL-C-9959, type III, shall be used.

5.2.1.4 Detached equipments. Detached voltage regulator and excitation components and accessories shall be preserved level A in accordance with MIL-S-196, MIL-R-2729 or MIL-E-17555, as applicable. Unless used in sets, pairs, or quantity greater than one as specified (see 6.2), detached components and accessories shall be individually preserved.

5.2.2 Commercial. Preservation of generator sets, detached components and accessories shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.3.1 Basic generator set.

5.3.1.1 Level A. Each unit shall be packed in a reusable plywood sheathed crate conforming to type II, class 2, style A or B of MIL-C-104. Each unit shall be anchored, blocked, and braced within the shipping container to resist shock and prevent damage during shipment and storage. Anchoring, blocking, and bracing shall be as specified in 5.3.3 and the appendix to MIL-C-104. Crates shall be provided with an inspection port for use in viewing the humidity indicator mounted in the barrier material. The rubbing strip of MIL-C-104 shall be 3-inch in lieu of 2-inch thick material.

5.3.1.2 Commercial. Packing shall be in accordance with ASTM D 3951.

5.3.2 Detached components and accessories. Detached components and accessories shall be packed level A or commercial as specified (see 6.2) in accordance with MIL-S-196, MIL-R-2729, or MIL-E-17555, as applicable. Rough handling test as specified in MIL-E-17555 shall not be required. Detached components and accessories shall be packed separately and secured in the engine shipping crate if space permits or shipped concurrently with the basic equipment.

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5.3.3 Anchoring, blocking, bracing, and cushioning. Anchoring, blocking, bracing, and cushioning of engine units, detached components and accessories shall be in accordance with MIL-STD-1186 and the applicable equipment or container specification and appendix thereto. In addition, cushioning materials, when used, shall be as follows:

(a) Use of loose excelsior or shredded newspaper is prohibited unless such are enclosed in a sealed, waterproof barrier conforming to PPP-B-1055.
(b) Use of loose-fill materials is prohibited for packing applications such as cushioning, filler, dunnage, and so forth.

5.4 Marking.

5.4.1 Standard markings. In addition to any special marking or instructions specified herein or by the contract or order (see 6.2), unit and intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4.2 Special markings. Shipping containers shall have the following marking:

(a) Adjacent to method II markings:

"STORE RIGHT SIDE UP - WARNING - SEE UNPACKING INSTRUCTIONS."

(b) Apply adjacent to the identification marking on the side of the container:

"CAUTION - THIS EQUIPMENT MAY BE SERIOUSLY DAMAGED UNLESS UNPACKING INSTRUCTIONS ARE CAREFULLY FOLLOWED. UNPACKING INSTRUCTIONS ARE LOCATED (state where located)."

(c) "REUSABLE INTERIOR AND EXTERIOR CONTAINER (when applicable)."
(d) "SEAL ALL ENGINE OPENINGS UPON REMOVAL OF BARRIER BAG."
(e) Handling and structural markings as applicable (see MIL-STD-129 and appendix to MIL-C-104."

5.5 Special instructions.

5.5.1 Preserved units. Generator sets preserved level A shall be tagged as follows:

"The fluid systems of this generator set have been preserved with grade 10 or 30 preservatives. No special depreservation procedures are required with the exception of the engine fresh water system. Depreservation for the fresh water system shall be in accordance with S9086-HD-STM-000."

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5.5.2 Depreservation. The contractor shall provide any additional depreservation instructions for systems or accessories when such instruction would not be applicable to 5.5.1 or would be necessary due to peculiarities of his equipment. Instructions shall be packaged in a transparent waterproof plastic bag, minimum 0.1 millimeters thick. Closure shall be by heat sealing. The shipping container in which the instructions are packed shall be marked to so indicate.

5.5.3 Unpacking. Unpacking instructions shall be provided for complex equipment or systems and floating bag type packs. The instructions shall contain the following information:

"To unpack remove the top and sides, leaving the unit resting on the bottom of the packing case. Remove the packing case and slip the unit off the base. In unpacking the item, the following precautions shall be observed to prevent possible damage:

(a) Observe the arrows marked on the shipping container. These point to the cover which can be removed most readily.
(b) Remove nails with a nail-puller only.
(c) Remove screws with a screwdriver only.
(d) Never pound or hammer the shipping container.
(e) Keep all levers and crowbars away from the interior of the container.
(f) Seal all engine openings upon removal of barrier bag."

5.5.3.1 Unpacking instructions. These instructions shall be placed in a sealed waterproof envelope prominently marked "UNPACKING INSTRUCTIONS" and firmly affixed to the outside of the shipping container in a protected location, preferably adjacent to the identification marking. If the instructions cover a set of equipment packed in multiple containers, the instructions shall be affixed to the number one container in the set or system.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Generator sets covered by this specification are intended for ship service and emergency service applications aboard surface ships or submarines.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Service category required (see 1.3).
(d) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) Whether first article test is required (see 3.1).
(f) Weight and dimensional limitations (see 3.3.1).

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(g) Mounting; whether generator is mounted off engine crank case; if not whether sub-base or ship foundation mounted (see 3.3.1.5).
(h) Whether set will be sound isolation mounted (see 3.3.1.5.5).
(i) Whether diesel engines conform to MIL-E-23457 or MIL-E-24455 (see 3.3.1.11).
(j) The direction of engine rotation if other than specified (see 3.3.1.11).
(k) Starting equipment required and operation of starting system (see 3.3.1.13.1).
(l) If electric starting, voltage and degree of enclosure for starting motor (see 3.3.1.13.1.1).
(m) If compressed air starting, pressure required for ship's air supply (see 3.3.1.13.1.2).
(n) Type of governing system (see 3.3.1.14).
(o) Whether a motor-operated speed changer shall not be provided for ac generator sets (see 3.3.1.14.1.1).
(p) Whether speed changer motors shall be other than as specified (see 3.3.1.14.1.1).
(q) Whether a centrifugal backup governor is required (see 3.3.1.14.2.3).
(r) Shock test if other than as specified (see 3.3.1.16.5 and 4.6.19).
(s) Short circuit requirements for type I sets (see 3.3.1.16.6.1 and 3.3.1.16.6.2).
(t) Design and special parallel operation requirements (see 3.3.1.16.7.1).
(u) When EMI control is not required (see 3.3.1.16.8 and 4.6.18).
(v) Power output ratings if other than specified.
(w) For type I sets only:

kW rating (see 3.3.2.1).
r/min (see 3.3.2.1).
Type of insulation (see 3.3.2.1).
Voltage (see 3.3.2.1).
If exciters, voltage regulators, field discharge resistors, battery charging generators, or starting batteries are required (see 3.3.2.2).
If generators operating in parallel require regulators or equalizers (see 3.3.2.5).
Mounting of balance coils (see 3.3.2.7).
If requirements for generator field rheostats are other than as specified (see 3.3.2.8).

(x) For type II and III sets only:

Rated capacity (see 3.3.3.1).
r/min (see 3.3.3.1).
Type of insulation (see 3.3.3.1).
If field discharge resistors, starting batteries, or switches are required for synchronizing motors (see 3.3.3.2).

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(y) For type IV sets only:

Rated snorkel and surface loads operation (see 3.3.5.2 and 3.3.5.2.1).
r/min (see 3.3.5.2).
Type of insulation (see 3.3.5.2).
Inlet vacuum and exhaust back pressures (see 3.3.5.2.1).
Horsepower, inrush current and p.f. of hydraulic pump motor (see 3.3.5.2.1(a)).
kW rating and inrush current of motor generator set (see 3.3.5.2.1(b)).

(z) List, pitch, roll and trim other than listed (see 3.3.5.3).
(aa) kW rating of the generator (see 3.3.5.5).
(bb) Submergence design and test pressures (see 3.3.5.5.3).
(cc) Voltage regulation and excitation type (see 3.3.5.6).
(dd) Mounting for static exciter and voltage regulator component enclosures (see 3.3.5.6.3).
(ee) Sound pressure levels if other than specified (see 3.3.7.1).
(ff) Structureborne and special airborne noise requirements, whether generator sets are to be resiliently mounted, and special design features not covered (see 3.3.7.2, 3.3.1.5.5, 3.3.7, 4.6.23.1 and 4.6.23.2).
(gg) Period for which on-line generator is to remain on-line (see 4.6.15.4).
(hh) Electromagnetic interference testing (see 4.6.18).
(ii) If test requirements of appendix A are not applicable (see 4.6.19).
(jj) If reliability demonstration test is required (see 4.6.20).
(kk) Navy fire-retardant material requirements (see 5.1.1).
(ll) Preservation requirements (see 5.2 and 5.2.1.3).
(mm) Quantity of detached components and accessories supplied (see 5.2.1.4).
(nn) Packing requirements (see 5.3 and 5.3.2).
(oo) Marking and special marking requirements (see 5.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailorin
3.3.1 and appendix B	DI-DRPR-80651	Engineering drawings	Level 3
3.3.1.2	DI-R-7082	Reliability prediction report	-----
	DI-R-7079	Reliability program plan	-----
3.3.1.2	DI-R-2129	Plan, maintainability demonstration	-----
3.3.1.2	DI-R-5189	Maintainability prediction data	-----
3.3.1.2	DI-MNTY-80822	Maintainability program plan	-----
3.3.1.2 and 4.6.21	DI-MNTY-80832	Maintainability/testability demonstration test report	-----
3.3.1.2.6	DI-R-7085	Failure mode, effects, and criticality analysis report	-----
3.3.1.3	DI-SAFT-80102	Safety assessment report	-----
3.3.1.16.4	UDI-S-26551	Report, vibration (self generated)	-----
3.3.1.16.4	UDI-T-23715	Analysis: torsional vibration for propulsion system (boats)	-----
3.3.1.16.8	DI-EMCS-80199	EMI control plan	-----
3.3.1.16.8	DI-EMCS-80201	EMI test plan	-----
3.3.1.16.8	DI-EMCS-80200	EMI test report	-----
4.3	DI-T-23731	Notification of tests	-----
4.4	DI-RELI-80939	Test and inspection report	-----
4.6.19 and appendix A	DI-T-2072	Reports, test	-----
4.6.23.1	DI-HFAC-80272	Equipment airborne sound measurements test report	-----
4.6.23.2	DI-HFAC-80274	Equipment structureborne vibration acceleration measurements test report	-----
4.7	DI-PACK-80120	Preservation and packaging data	-----
4.7	DI-PACK-80121	Special packaging instructions (SPI)	-----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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6.4.1 Manuals with shipment. Technical manuals which accompany shipments should be packaged in a transparent waterproof plastic bag, minimum four mil thick. Closure should be by heat sealing. Technical manuals should not be placed with any flexible sealed barrier enclosing components. The copy(s) of the manual should be placed in the shipping container housing the main unit. Packing lists should indicate which container contains the technical manual(s) and should state the approximate location therein. The manual should be readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, should not be individually wrapped, but should be packed in accordance with the requirements of the applicable technical manual specifications.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

6.6 Resilient mount submittal. Where resilient mounts are used, the following information should be submitted to the activity acquiring the generator set for review:

(a) Speeds of the equipment involved.
(b) Total weight of the mounted assembly in the operating condition. The weight shall include bedplate, service fluids, piping, filters, and all other attached accessories.
(c) Location of the center-of-gravity of the assembly in the operating condition in three planes.
(d) The moments of inertia and products of inertia of the assembly about three mutually perpendicular axis with the origin at the assembly center of gravity of the mounted unit and the orientation of the axis indicated with respect to the equipment.
(e) The six rigid body natural frequencies of the mounted assembly in the operating condition.
(f) The type of resilient mount used and any assumptions made in calculating the natural frequencies.

6.7 Personnel skill levels. The training and skill levels of the contractor's shop personnel who perform the maintenance tasks should be commensurate with Navy personnel who will maintain the generator set in the judgement of the contractor and with the approval of NAVSEA (see 4.6.21.1).

6.7.1 Simulation of faults or failures. Simulation of faults or failures the introduction of defective parts is not required.

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6.8 Material safety data sheet (MSDS). Contracting officers must identify those activities requiring copies of MSDS's. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.9 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.10.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.11 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.12 Type designations. Type designations (see 1.2) have been changed. Previous designations were as follows:

MIL-G-3048
Type I - dc
Type II - ac

MIL-G-21296
Type I - Single engine driven, ac
Type II - Twin engine driven, ac

6.13 Subject term (key word) listing.

Air coolers
Balance coils
Centrifugal governor
Contacts
Hydraulic system
Load sensing
Power rectifiers
Speed sensing
Terminals

6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6115-N530)

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APPENDIX A

REQUIREMENTS FOR HIGH IMPACT (H.I.) SHOCK EQUIPMENT

10. SCOPE

10.1 Scope. This appendix covers the design, test, and acceptance requirements for diesel generator sets which must withstand high-impact shock. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-21952 - Steel (HY-80 and HY-100) Bars, Alloy.
MIL-S-22664 - Steel, Structural Shapes Alloy, High Yield Strength (HY-80 and HY-100). (Metric)
MIL-S-23008 - Steel Castings, Alloy, High Yield Strength (HY-80 and HY-100).
MIL-S-23009 - Steel Forgings, Alloy, High Yield Strength (HY-80 and HY-100).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. REQUIREMENTS

30.1 Requirements for high-impact shock equipment. The requirements shall be in accordance with MIL-S-901 except as otherwise specified herein.

30.2 Specific shock requirements. Diesel generator sets and all attached and detached accessories shall conform to the requirements for type A, class I of MIL-S-901 and the shock test criteria specified in 30.4. Acceptance of generator sets and accessories which have been subjected to shock tests shall be based on the ability of the equipment to conform to acceptability criteria, as revealed by

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its performance during and after shock tests, and the results of post-test examination and operating tests conducted at the contractor's plant. In the event that the acceptance criteria are not met, acceptance shall also depend upon the corrective action proposed by the contractor. The contractor's proposals for shock hardening (if necessary) shall be forwarded to NAVSEA. Review by NAVSEA of the results of shock tests, pre-shock and post-shock test inspections, and corrective action shall be required.

30.3 Function. Shockproof generator sets shall continuously deliver electrical power conforming to required performance characteristics during and after being subjected to shock of the magnitude experienced during shock tests as specified in MIL-S-901. A shockproof generator set, when subjected to shock, shall not suffer damage to the extent that it creates a possible hazardous situation such as fire or injury to operating personnel, or such as to result in likely failure of major units of the set if the set is not immediately shut down for corrective action. In determining if equipment has passed shock tests, the criteria specified in 30.4 shall be used.

30.4 Acceptability criteria for shock tested equipment. In determining if a shock tested equipment has passed shock tests, the criteria specified in 30.4.1 through 30.4.2.6 shall apply. NAVSEA shall be the final judge of whether an equipment shall be accepted or rejected on the basis of shock test requirements; however, the contractor shall have the option of convening a conference with NAVSEA to discuss questionable situations involving acceptance or rejection of shock tested equipment.

30.4.1 Damage. Damage as specified below shall not be permitted as a result of shock tests:

(a) Damage which causes or necessitates reduction of equipment output to a level below rated for any length of time within 30 days after attack.
(b) Damage that renders inoperative any item or device installed to protect the equipment or operating personnel, such as speed limiting governors, overspeed trips, low r/min shutdown devices, and high temperature low lube oil pressure alarms.
(c) Shock-induced actuation of safety devices specified in (b).
(d) Damage or shock-induced misadjustment of speed governing systems that results in inability to maintain speed automatically within the limits imposed by the individual equipment specifications until repairs or adjustments are made.
(e) Loosening or shifting of any item affixed to a rotating part.
(f) Damage that causes any assembly or subassembly to come adrift from any of its intended restraints.
(g) Damage that requires repairs of any type to be made before equipment operating parameters can be adjusted to suit operational requirements.
(h) Damage that creates a safety hazard to personnel or equipment.
(i) Complete fracture of any unit or welded assembly.
(j) Complete stripping or complete failure by gross distortion of all engaged threads (male or female) of any threaded part.

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(k) Increase of dynamic unbalance of any rotating part or assembly to such an extent that the vendor must rebalance the affected item prior to delivery of the equipment or prior to conducting field balancing operations.

30.4.2 Damage for items. Damage from shock tests as specified herein shall be permitted to the extent or under the conditions listed for each item.

30.4.2.1 Damage to wearing parts. Damage that shortens the useful life of wearing parts by initiating or aggravating some long-term detrimental effect such as wear, fatigue, or erosion is permissible if the useful life of wearing parts is reduced by no more than 25 percent. Due to difficulties associated with estimating reduction of useful life, equipment acceptability shall be determined with consideration for the accuracy of the estimate. In cases where it is especially difficult to estimate percent reduction of useful life, equipment will not be rejected for this cause unless, in the opinion of NAVSEA, the limits specified herein have been grossly exceeded. In cases where the availability of test or maintenance information permits a more accurate estimate of useful life reduction to be made, appropriately less leniency will be allowed when determining acceptability of the equipment on this basis.

30.4.2.2 Damage resulting in minor leaks. Damage resulting in minor leaks of fluids or gases is permissible if equipment still meets the requirements of this appendix.

30.4.2.3 Loosening of hold-down. Loosening of hold-down is permissible if equipment still meets the requirements of this appendix.

30.4.2.4 Damage that requires adjustment. Damage that brings about a requirement to adjust control apparatus or equipment setpoints in order to regain the pre-attack mode of operation, or damage that brings about a requirement for timely repairs or compensatory action in order to prevent further (and eventually unacceptable) damage that is related to increased wear rates or other long term effects, is permissible only if the following conditions are met:

(a) If NAVSEA agrees that the damage or its effects are of such a nature that timely and proper damage diagnosis and compensation would certainly be made by ship's force personnel familiar with the equipment. (For purposes of this appendix, it is considered that misalignment (if any) of shafting cannot be corrected by ship's force.)
(b) If the equipment status before adjustment or repair is made is such that equipment still meets the requirements of this appendix.

30.4.2.5 Damage that results in rubbing. Damage that results in continuous or intermittent rubbing contact between stationary and moving parts (during equipment operation) in areas where it was not originally intended that such contact be made generally shall not be permitted. NAVSEA may choose to permit, on a case basis, rubs which quickly wear away without causing adverse effects which cause the equipment to fail to meet the requirements of this appendix.

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30.4.2.6 Damage requiring moderate increase of labor before repairs are made. Damage that brings about (until repairs are made) a requirement for slight to moderate increase of manual effort, maintenance, or attention on the part of equipment operators in order to maintain the operating mode or the machinery space environment shall be permissible when specifically agreed to by NAVSEA.

30.5 Construction.

30.5.1 Hold-down bolts. The generator set hold-down bolt holes provided for securing equipment to the ship's foundation shall be constructed for use with nonfitted bolts on one side and fitted bolts on the other. Maximum clearance between bolt holes and bolts shall not exceed the following:

Nominal bolt diameter	Maximum diameter of hole
3/4 inch or smaller	Nominal bolt diameter plus 1/32 inch
Larger than 3/4 inch	Nominal bolt diameter plus 1/16 inch

30.5.2 High yield strength materials. When high yield strength materials (materials with yield strength of a gauge pressure of 75,000 lb/in² or more) are used, by reason of choice or requirement, to meet design stress levels, or to satisfy resistance to shock, or both, HY-80 or HY-100 in accordance with MIL-S-21952, MIL-S-22664, MIL-S-23008, or MIL-S-23009, as applicable, shall be used.

40. QUALITY ASSURANCE PROVISIONS

40.1 General inspection. Complete generator sets and detached accessories shall be subjected to and successfully pass shock tests.

40.2 Test requirements. Shock tests shall be conducted in accordance with the grade A machinery requirements and type A tests of MIL-S-901 and as follows:

(a) The generator set package (prime mover, generator, and all attached accessories) and any detached accessories needed to control and operate the set shall be shock tested as a principal unit if the weight size and arrangement limitations are not exceeded. Detached accessories not tested with the generator set package shall be individually tested using the applicable shock testing machine referenced in MIL-S-901 and in accordance with the individual equipment specification.

(b) Assemblies or subassemblies comprising the equipment to be shock tested shall be those actually made for and furnished under the contract

(c) Assemblies and subassemblies comprising the equipment furnished for shock tests shall comply with the individual equipment specification requirements. Required performance tests shall have been completed before shock testing. Request for exceptions requirements will be considered on a case basis by NAVSEA.

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(d) The contractor shall carefully inspect equipment which will be shock tested in order to detect any flaws or undesirable conditions that might contribute to shock test damage, or be mistaken for shock damage during the post-test inspection. For purposes of this appendix, damage is defined as any permanent change of the status or condition of any equipment that results from shock testing.

(e) Running clearances, alignments, runouts, and mechanism set points that are normally established within the equipment and its ancillary devices by the contractor shall be properly established in the equipment before shock tests, even though it may be the contractor's usual practice to establish certain of these settings after installation of the equipment onboard ship. If the contractor desires, setpoints may be established by setting or adjusting the device in question in such manner that the correct setpoints are theoretically achieved (that is achieved without conducting actual tests). Setpoints define the input-output relationships of any mechanisms or device associated with the equipment to be shock tested. For instance, the setpoint or an overspeed tripping device could be actuation (output) at 10,000 r/min (input).

(f) If setpoints are obtained in the manner specified in (e), however, the contractor shall document the pre-test status of all adjustments, dimensions, and characteristics that could possibly cause change of setpoints as a result of shock test damage. If the contractor desires that certain alignments, clearances, or setpoints be established by the shock testing facility as the equipment is erected for shock tests (necessary, for instance, if equipment is disassembled for shipment), the cognizant shock testing facility shall be so informed at least 2 months before the equipment is to be delivered to the shock testing facility. Due to facility limitations, the contractor may be required to provide personnel or equipment, or both, necessary to accomplish such work.

40.3 Equipment mounting during tests.

40.3.1 Heavy weight equipment. Shock tests shall be conducted on the floating shock test platform specified in MIL-S-901.

40.3.1.1 Mounting. When equipment is expected to be used in various shipboard applications involving different locations, the generator set package shall be mounted on a typical ship's foundation welded to the inner deck of the test barge. Equipment intended for one particular application only may be mounted on a foundation simulating that involved in the installation. Deck mounted detached accessories, if tested with the generator set package, shall be mounted on a fixture that accommodates the accessory and is welded to the inner deck.

40.4 Test procedure. Shock tests shall be conducted in accordance with MIL-S-901 and as specified in 40.4.1 and 40.4.2.

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40.4.1 Procedure. Equipment shall be operated or energized during each blow or shot except for the first blow or shot which may be performed unenergized and not operating. Before each blow or shot, however, the generator set shall be run at rated speed and voltage, no-load, for a period of time necessary to determine that all components are functioning normally. Components and accessories, attached or detached, which contain a fluid during normal operation, (such as coolers, and so forth) shall contain a fluid of comparable weight during all blows. The equipment shall be examined for damage after each blow. Correction of minor damage during the progress of tests will be permitted and the test may proceed to the next level of shock after the minor damage is corrected. Major damage shall be cause for rejection and suspension of tests until such time as the contractor's proposed corrective action is approved by the Command or agency concerned. In such cases, after corrective action is completed, re-test shall be required starting at the test level which caused the damage. The decision as to whether damage is minor or major shall be made by the representative of the Command or agency concerned at the test site. In case of disagreement, the matter shall be referred to the Command or agency concerned for final decision.

40.4.2 Instrumentation and measurements. Instrumentation and measurements during the tests shall be as follows:

(a) Velocity pickups, high speed movie cameras, instrumentation for monitoring set operation, and provision for remote shutdown shall be installed.
(b) Initial start-up and check-out shall be conducted under guidance the contractor.
(c) During the last blow or shot, an oscillograph recording shall be made showing generator output voltage and frequency characteristics from a time beginning with 5 seconds before the blow until 30 seconds after the blow.
(d) Scribe lines or punch marks shall be used to indicate exact location of all major attached accessories and components (such as blowers, pumps, generator feet, end covers, and so forth) before shock tests so that any relative motion during shock blows can be measured and recorded.
(e) Generator shaft run-out shall be measured before starting test and after each blow.
(f) Air-gap measurements shall be taken before start of tests and after each blow.
(g) Crankshaft deflection shall be checked before start of test and after each blow, if feasible (measurement to be made by the contractor's representative).
(h) With the set operating at rated speed, first order vibration levels shall be measured with a vibration pick-up in the X, Y, and Z directions before and after each shot.
(i) Insulation resistance and dielectric strength shall be measured before start of testing and after each blow.
(j) A thorough visual examination shall be made after each blow and any indications of movement or distortion recorded.

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(k) Generator cooling covers shall be removed after each shot for visual examination of the cooler and any excitation parts.
(l) Provision shall be made for remote observation of lubricating oil pressure to bearings and for remote shutdown of the equipment during the last shot.
(m) Additional instrumentation shall be provided by the testing activity as required.

40.5 Examination for damage. After completion of shock tests, the equipment shall be returned to the contractor's plant for a complete teardown and examination for damage. Upon receipt in the contractor's plant, before removal from the carrier, the equipment shall be given a visual examination to assure that no damage has occurred during transit. The equipment shall then be removed to the test floor area and all examination covers of all components removed and the components shall be examined to determine runability of the set. If the examination reveals no damage which would prohibit full power operation, the set shall be installed on a test stand and operated at full load for a minimum of 4 hours. Engine and electrical equipment performance shall be monitored and recorded during the load run. Safety devices, such as overspeed trip, low r/min trip, and so forth, shall be tested in an appropriate manner three times to determine if they are operating properly at the correct set points. Upon completion of the tests, the set components shall be disassembled part by part. Visual examination and dimensional checks of parts and clearances shall be made as necessary, and damage, evidence of shifting, rubbing, momentary contacts and so forth shall be recorded. Piping and coolers shall be given hydrostatic tests for evidence of any leaks.

40.6 Disposition of shock tested equipment. Unless otherwise specified in individual equipment specifications, shock tested equipment will be acceptable provided the following requirements are met:

(a) Damage resulting from shock tests is corrected. The method of correction shall be approved by the Command or agency concerned.
(b) Bearings shall be replaced where examination reveals obvious but minor damage. Where damage to bearings is discovered, appropriate corrective action shall be taken.
(c) Any improvements necessary to shock harden are incorporated in the unit.
(d) The reworked unit is given the duplicate unit tests of this specification and the applicable individual equipment specification; the results are consistent with results of tests taken on other identical units.

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50. ARRANGEMENTS

50.1 Light weight and medium weight tests of accessory items (if used). Shock tests utilizing the light weight or medium weight shock test machines may be made at any commercial or Government activity having an approved test machine. The contractor may make arrangements for such tests direct with the activity of his choice. A list of such activities having approved shock test machines in the area of the contractor's plant may be obtained by written request to the Command or agency concerned.

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APPENDIX B

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements for a master drawing to be furnished for each diesel generator (DG) set when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENTS

30.1 Drawing contents. A master drawing shall be included and shall include as a minimum the following.

30.1.1 Engine and generator data in a format typically as shown on figure 1.

30.1.2 An assembly end and side view, showing all parts of the DG set. The assembly side view shall show a longitudinal section of the DG set above the centerline. The assembly end and view shall show a transverse quarter section above the centerline. The following minimum information shall be shown.

30.1.2.1 Overall dimensions for components and the complete generator set, foundation bolting dimensions, wet and dry weights, including detached accessories, center-of-gravity in at least two views, clearances for removal of parts such as cylinder heads, pistons, liners, cooler tubes, and generator rotor.

30.1.2.2 Data necessary to clearly describe all equipment, including capacities of blowers and pumps at rated speed, volume of exhaust gases at rated speed, rated brake (bhp), brake mean effective pressure (bmep), r/min, type numbers, ratings (volts, amperes, r/min, and kW manufacturer).

30.1.2.3 Location and size of pipe connections to engine, gear, and accessories, electrical connections.

30.1.2.4 Diagrammatic sketches of piping systems, indicating pipe sizes, types of connections, relative locations of coolers, filters, valves, strainers, thermometers, gauges, quantity and temperatures oil and water. Manufacturer furnished piping shall be shown by solid lines, and shipbuilder furnished piping by broken lines.

30.1.2.5 An outline of the identification plate showing all plate markings and the location of the plate.

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30.1.2.6 The moments of inertia and products of inertia of the assembly about three mutually perpendicular axes with the origin at the center-of-gravity of the sound mounted unit and the orientation of the axis indicated with respect to the equipment.

30.1.2.7 The six natural frequencies (in Hz) of the assembly in the operating condition.

30.1.2.8 Combustion air consumption in cubic feet per minute (ft^3/min) under the standard operating conditions specified for the engine.

30.1.2.9 Heat dissipation to the space in British thermal unit per hour (Btu/hr) at full load.

30.1.2.10 Alarm contact devices (as applicable):

(a) Air temperature alarm contact calibrated __°F to __°F, set at __°F.
(b) Bearing temperature alarm contact calibrated __°F to __°F, set at __°F.

30.1.3 Master drawing covering equipment which has been shock-tested and found to be satisfactory shall be so indicated on the drawing by the following: "This generator has satisfactorily passed shock tests as recorded in (activity) test report number _____ forwarded to the bureau or agency concerned via Government inspection _____ letter _____, dated _____."

30.1.4 A space for Government inspector's signature, office, and date.

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ENGINE DATA		
ENGINE B.M.E.P		
PUMP CAPACITIES	R.P.M.	GPM TOTAL HEAD IN FT.
FRESH WATER		
SEA WATER		
LUB OIL SUPPLY		
FUEL OIL		
BLOWER PRESSURE RATIO		
BLOWER		
ENGINE CYL. DIS.		
COOLANT CAPACITY		GALLONS
LUBE OIL SUMP CAPACITY WHEN RUNNING		GALLONS

GENERATOR DATA				
GENERATOR ROTOR WEIGHT WITH SHAFT				LBS EST
GENERATOR COOLER WEIGHT	DRY	LBS.	WET	LBS
GENERATOR COOLER REQUIRES	G.P.M. COOLING WATER			
PRESSURE DROP THRU AIR COOLER	INCHES OF WATER - STATIC			
REFERENCE DRG. DATA ETC.				

ENGINE COMPONENTS	WEIGHTS
TOP COVER	LBS.
UPPER CRANKSHAFT	LBS
PISTON WITH PIN AND INSERT	LBS
CONNECTING ROD UPPER	LBS
CONNECTING ROD LOWER	LBS
CONNECTING ROD BEARING SHELL	LBS
CYLINDER LINER AND JACKET	LBS
CYLINDER LINER AND JACKET WITH EXHAUST BELT	LBS
BLOWER	LBS
EXHAUST MANIFOLDS (EACH SIDE)	LBS

WEIGHTS	DRY	WET
ENGINE COMPLETE		
GENERATOR		
SUBBASE		
COMBINED UNIT		
DETACHED ACCESSORIES		

UNIT	MFG	DRG. NO.	BUSHIPS NO.
GENERATOR			
LIST OF ENGINE PLANS			
LIST OF INST. PLANS			

FIGURE 1. Typical format for engine and generator data.

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